Independent variable: x-axis
dependent variable: y-axis
Keep everything else constant other than “determine the value of, independ, depend”

Diagram:

Clamp, stand, bench, micrometer

Method of data collection (measure):

Use top-pan balance to measure mass
newton-meter to measure weight of load
set-square to check that clamped ruler is vertical
micrometer to determine diameter d of wire
timer/stop-watch
oscilloscope for “few milliseconds”
thermometer for temperature. (final – initial) then change in temp
ruler
protractor to measure θ

Adhesive putty, tape so it sticks

To find Area of cross section, determine diameter using micrometer then use $A = (\pi D^2) / 4$ or A = length x width
To find density: mass(balance)/volume(measuring cylinder)
Surface area of cylinder: $\pi dL + \pi d^2/2$

Method of analysis: 3-4 marks guaranteed if you write these: (You wont get a mark for writing just the gradient and y-intercept)

Only after making it into y = mx + c then only (Cannot just copy it from the question): Plot a graph of “y-axis” against “x-axis” (Plan a laboratory experiment to test the relationship between **y** and **x**) – NEED TO DRAW THE GRAPH

Gradient = “”, y-intercept = “”

Explain how the results could be used to determine values for “U” and “Z” means make U and Z the subject according the equation given. (Scroll down Example 1 and example 2 is there)

(0,0) or y = mx: relationship is valid if a straight line is produced (passing through the origin).
If there is y-intercept: relationship is valid if a straight line is produced (not passing through the origin).

Safety precaution:

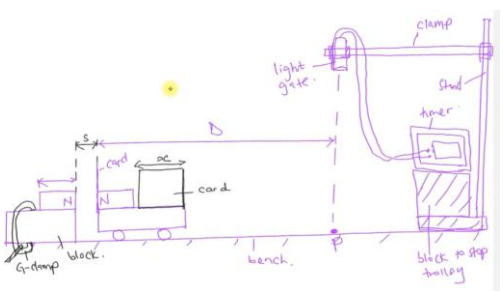
Eye googles
Safety screen to prevent eye injury from moving spring
Use cushion/sand box in case load falls
Gloves to prevent oil contacting skin or Gloves to prevent electric shock or Gloves to prevent burns from hot
Wear ear defenders to prevent damage to hearing

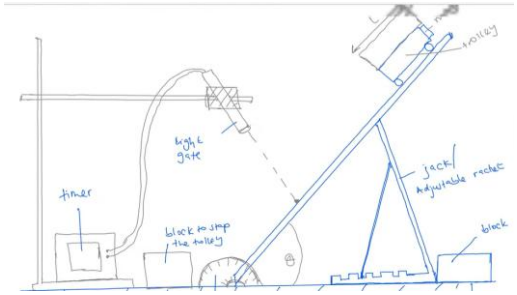
Use insulating gloves to prevent shocks from high voltage

Voltage – Voltmeter
Current – Ammeter
Resistance – Ohmmeter

Ensure fully submerged
Stir the water using glass rod

Light gate:
Light gate connected to data logger





- Place a block to stop the trolley

Wavelength of light:
- Wear goggles to protect eyes from light
identify the zero order and count the number of maxima.
Use intense light source and perform in a dark room.
To determine λ: Using diffraction grating by measuring the angle θ of the bright maxima and using d sinθ = nλ

Magnetic flux density:
Measure B using hall probe and adjust probe until maximum value.

Frequency:
Use CRO (cathode ray oscilloscope) to measure time period
To determine frequency from oscilloscope, period from oscilloscope (time period) = time-base x horizontal distance and then use frequency = 1 / time period

D6	method to measure *f*, e.g. read from signal generator or use of oscilloscope
D7	method to determine *f* from oscilloscope, e.g. period from oscilloscope T = time-base × horizontal distance **and** $f = 1/T$

“Plan a laboratory experiment to test the relationship between **t** and **θ**” at the end you can write “repeat measurements of **t** for the same **θ** and average **t**”
“Plan a laboratory experiment to test the relationship between **s** and **A**” at the end you can write “repeat experiment for each **A** and average **s**”

Repeat measurement of diameter of different “” and average.

Example 1: method of analysis for 2nd and 3rd mark

It is suggested that the relationship is

$$ma = F - (mg\sin\theta + k)$$

where g is the acceleration of free fall, m is the mass of the trolley and k is a constant.

Design a laboratory experiment to test the relationship between a and θ. Explain how your results could be used to determine a value for k. You should draw a diagram, on page 3, showing the

1. Make it into y = mx + c form, find gradient and y-intercept. “a” is y and “θ” is x
2. After finding y-intercept / gradient make “k” subject

Example 2: method of analysis for 2nd and 3rd mark

It is suggested that E is related to R by the relationship

$$E = 2\pi fM\left(\frac{V}{R + k}\right)$$

where V is the potential difference across the resistor and coil P, and k and M are constants.

Plan a laboratory experiment to test the relationship between E and R.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for k and M.

1. Make in the form of y = mx + c. Take reciprocal first since the “x-axis (R)” is on the denominator
2. After you find the gradient and y-intercept -> using these make k and M the subject
3. One of them has to be known to make other variable the subject. Example you made M the subject, not necessary to make M the subject for other equation.

Eg: M = 1 / grad x 2 pi f v

K = y-int x 2 pi f M v, here u cant make M subject cuz k is unknown. We made k subject cuz M is known from 1st equation